

Fashion Magazine Generator

Step-by-step guide to install and run the app on macOS and Windows

The Fashion Magazine Generator is a web app that runs on your own computer. You describe an article and the images you want, and the app writes the article, generates the images, checks them with CLIP and assembles everything into a magazine page. This guide takes you from an empty computer to a running app. Follow the steps in order and do not skip any.

What you need

- A Mac with an **Apple Silicon chip** (M1, M2, M3 or M4), or a **64-bit Windows 10 or 11** PC.
- About **3 GB of free disk space** and a stable internet connection.
- Your **OpenRouter API key**. It starts with `sk-or-`. If you do not have one, ask a Teaching Assistant.
- About **30 minutes** for the first setup. Starting the app again later takes about one minute.

How to read this guide

Every step that differs between the two systems has a **macOS** part and a **Windows** part. Only follow the part for your computer.

Text in a blue box is a command. Type it (or copy and paste it) into the terminal exactly as shown and press **Enter**. Run the commands of a box one line at a time, in order.

You do not need the files `setup.sh`, `setup.ps1` or `instructions.pdf` from the project folder. This guide replaces them.

Overview

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4	Create and activate a virtual environment
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8	Give feedback and edit images
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1 Check your computer and open a terminal

Check your computer

macOS

1. Click the **Apple menu** in the top left corner of the screen and choose **About This Mac**.
2. Look at the line **Chip**. If it says **Apple M1, M2, M3** or **M4**, your Mac is supported.
3. If you see **Processor** with **Intel** instead, your Mac cannot run this app, because the AI library it uses (PyTorch) no longer supports Intel Macs. Use the web app at the end of the Colab notebook instead, or ask a Teaching Assistant.

Windows

1. Press the **Windows key**, type **About your PC** and press **Enter**.
2. Under **Device specifications**, check that **System type** says **64-bit operating system, x64-based processor**.

Open a terminal

The terminal is a window where you type commands. Keep it open during the whole setup.

macOS

1. Press **Cmd + Space** to open Spotlight.
2. Type **Terminal** and press **Enter**.

Windows

1. Press the **Windows key** and type **PowerShell**.
2. Click **Windows PowerShell**. You do not need to run it as administrator.

Important for Windows users

All Windows commands in this guide are written for **PowerShell**. Do not use the older **Command Prompt** (cmd.exe), because some commands are different there.

2 Install Python 3.13

The app needs Python 3.13. First check if it is already installed. If the check succeeds, skip to Step 3.

Check if Python 3.13 is installed

macOS

```
python3.13 --version
```

Windows

```
py -3.13 --version
```

If the answer is **Python 3.13.** followed by a number (for example **Python 3.13.2**), Python is installed. Go to Step 3. If you see an error such as **command not found, is not recognized** or **No suitable Python runtime found**, install Python as described below.

Install Python on macOS

1. Open <https://www.python.org/downloads/macos/> in your browser.
2. Under **Stable Releases**, find the newest **Python 3.13** release (for example Python 3.13.9) and click **Download macOS 64-bit universal2 installer**. Do not pick Python 3.14 or newer.
3. Open the downloaded **.pkg** file from your Downloads folder. Click **Continue** on every screen, click **Agree**, then **Install**, and enter your Mac password when asked.
4. When the installation finishes, a Finder window with the folder **Python 3.13** opens. Double-click the file **Install Certificates.command** in that folder. A terminal window opens, runs for a few seconds and prints **update complete**. You can then close that window. This step lets Python download files securely.
5. Close your terminal window completely (**Cmd + Q**), open a new one as in Step 1, and run the check again:

macOS

```
python3.13 --version
```

Expected answer: Python 3.13 followed by a number.

Install Python on Windows

1. Open <https://www.python.org/downloads/windows/> in your browser.
2. Under **Stable Releases**, find the newest **Python 3.13** release and click **Download Windows installer (64-bit)**. Do not pick Python 3.14 or newer.
3. Open the downloaded **.exe** file. On the first screen, tick the box **Add python.exe to PATH** at the bottom of the window. Then click **Install Now** and click **Yes** if Windows asks for permission.
4. On the last screen, if you see the option **Disable path length limit**, click it and confirm with **Yes**. Then click **Close**.
5. Close PowerShell, open a new PowerShell window as in Step 1, and run the check again:

Windows

```
py -3.13 --version
```

Expected answer: Python 3.13 followed by a number.

3 Download the project files

All files are in the public GitHub repository **eth-bmai-hs26/w3-project-public**. You will place the project in your **home folder**, so that every command in this guide works without changes. Choose **Option A** if you have never used Git.

Option A: download a ZIP file (recommended)

1. Open **<https://github.com/eth-bmai-hs26/w3-project-public>** in your browser.
2. Click the green **Code** button, then click **Download ZIP**. The file **w3-project-public-main.zip** is saved in your Downloads folder.
3. Unzip it. **macOS**: double-click the ZIP file. **Windows**: right-click the ZIP file, choose **Extract All**, then click **Extract**.
4. Open the extracted folder **w3-project-public-main**. On Windows it often contains a second folder with the same name. Open folders until you see **fashion_app**, **notebook** and **README.md** together. The folder that contains these three is the project folder.
5. Rename the project folder to exactly **w3-project-public** (without -main).
6. Move it into your home folder. **macOS**: in Finder press **Shift + Cmd + H** to open your home folder and drag the project folder into it. **Windows**: in File Explorer, click the address bar, type **%USERPROFILE%**, press **Enter**, and drag the project folder into that window.

Windows: why not the Documents folder?

On many Windows PCs the Documents folder is synchronised with OneDrive. OneDrive then tries to upload thousands of files from the virtual environment, which slows down your PC and can break the installation. Your home folder is not synchronised.

Option B: clone with Git

Only use this if Git is already installed on your computer.

macOS

```
cd ~
git clone https://github.com/eth-bmai-hs26/w3-project-public.git
```

Windows

```
cd $HOME
git clone https://github.com/eth-bmai-hs26/w3-project-public.git
```

Check that the files are in the right place

Go to the app folder and list its contents:

macOS

```
cd ~/w3-project-public/fashion_app  
ls
```

Windows

```
cd $HOME\w3-project-public\fashion_app  
dir
```

You should see, among others, `backend`, `Data`, `templates` and `requirements.txt`. If you get an error such as **No such file or directory** or **Cannot find path**, the folder is not in your home folder or has a different name. Repeat this step.

4 Create and activate a virtual environment

A virtual environment is a private folder named `venv` that holds all the packages of this project, separate from the rest of your computer. You create it once.

Create the virtual environment (only once)

Make sure you are in the `fashion_app` folder, then create the environment:

macOS

```
cd ~/w3-project-public/fashion_app  
python3.13 -m venv venv
```

Windows

```
cd $HOME\w3-project-public\fashion_app  
py -3.13 -m venv venv
```

The command prints nothing when it succeeds. A new folder `venv` now exists inside `fashion_app`.

Activate the virtual environment (every time you open a new terminal)

macOS

```
source venv/bin/activate
```

Windows

```
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass -Force  
.\venv\Scripts\Activate.ps1
```

The first line allows PowerShell to run the activation script in this window only. It does not change any setting of your computer.

When the environment is active, your prompt starts with `(venv)`. Check that the right Python is used. From now on the command is simply `python` on both systems:

Both

```
python --version
```

Expected answer: Python 3.13 followed by a number.

Always check for (venv)

Every command from Step 5 onwards must run with the environment active. If `(venv)` is missing at the start of the prompt, run the activation command above again from the `fashion_app` folder.

5 Install the required packages

With `(venv)` visible and inside the `fashion_app` folder, run these two commands one after the other. They are the same on macOS and Windows:

Both

```
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
```

The second command downloads and installs about 75 packages, including PyTorch, Flask, OpenCV and Transformers. It downloads between 300 MB and 1 GB and usually takes **5 to 15 minutes**, depending on your internet connection. Lots of text scrolls by; this is normal. Do not close the terminal while it runs.

The installation succeeded when the last lines start with **Successfully installed** and you get the prompt back without a red **ERROR** line. A yellow **notice** or **WARNING** line is not a problem.

If the installation stops with an error

The most common message is **No matching distribution found for torch==2.10.0**. It means the virtual environment was created with the wrong Python version, or you are on an Intel Mac. Check `python --version` inside the environment. If it does not show Python 3.13, delete the `venv` folder and repeat Step 4 with Python 3.13. See Step 10 for more problems and fixes.

6 Start the app and open it in your browser

The app must be started from the `backend` folder inside `fashion_app`. With `(venv)` visible, run:

macOS

```
cd ~/w3-project-public/fashion_app/backend
python main.py
```

Windows

```
cd $HOME\w3-project-public\fashion_app\backend
python main.py
```

What you will see

- The **first start takes 2 to 5 minutes**, because the app downloads two AI models (about 700 MB) that it keeps for later. Later starts take about one minute.

- You see loading messages such as **[BERT] Model ready.** and **fashion-CLIP loaded.** They may appear twice. This is normal.
- When the app is ready, the terminal shows **Running on http://127.0.0.1:5001.**
- A red line **WARNING: This is a development server** is also normal and can be ignored.

Open the app

1. Open a web browser (Chrome, Firefox, Edge or Safari).
2. Type **http://127.0.0.1:5001** into the address bar and press **Enter**.
3. The page **Fashion Generator** opens, with a form on the left and the text **Your article will appear here** on the right.

Keep the terminal open

The terminal runs the app. If you close it, the app stops and the page no longer works. Leave it open while you use the app.

7 Enter your API key and generate your first article

Save your OpenRouter API key

1. At the top of the left sidebar, click into the field **OpenRouter API Key**.
2. Paste your key. It starts with **sk-or-v1-**. The key stays hidden while you type; click the eye symbol next to the field to show it.
3. Click **Save Key**. The app checks the key with OpenRouter, which does not use any credits.
4. When the key is accepted, the text **Key saved for this session** appears below the button.

The key is kept in memory only. After you stop and restart the app you enter it again, or you store it in a file as described below.

Optional: store the key in a .env file

This saves you from entering the key after every restart. Open a **second** terminal window (the first one is running the app), go to the **fashion_app** folder (not **backend**) and run the command for your system. Replace **sk-or-v1-your-key** with your real key and keep the quotes.

macOS

```
cd ~/w3-project-public/fashion_app
echo "OPENROUTER_API_KEY=sk-or-v1-your-key" > .env
```

Windows

```
cd $HOME\w3-project-public\fashion_app
Set-Content -Path .env -Value "OPENROUTER_API_KEY=sk-or-v1-your-key" -Encoding ascii
```

The key is used the next time you start the app (Step 9). Never share this file and never upload it to GitHub, because it contains your private key.

Generate an article

1. **Article Title:** enter the title, for example *Spring Florals at ETH*.

2. **Image Styles:** describe the first image, for example *students in floral dresses, golden hour, blurred background*. Each field creates one image. Click **+ Add image style** to add a field for another image, or the **×** button next to a field to remove it.
3. **Article Description:** describe in a few sentences what the article should cover. The app uses this text to search its fashion dataset and to write the article.
4. Click **Generate**.

The sidebar shows the progress live: refining your input, retrieving fashion data, generating each image with its CLIP score, writing the article and assembling the page. A full run usually takes **1 to 3 minutes**. When it is finished, the article appears on the right.

Good to know

- Every image and every article costs OpenRouter credits. Avoid clicking Generate repeatedly while a run is in progress.
- All articles and images are saved in the folder `fashion_app/output` on your computer.
- Click **Session History** at the bottom of the sidebar to see your previous articles. **Delete History** clears this list. It does not delete the files in `output`.

8 Give feedback and edit images

After the first article, a feedback section appears in the sidebar.

Change the text or regenerate images

1. **Article Text Changes:** describe what should change in the text, for example *Make the introduction shorter*.
2. **Image 1, Image 2, ...:** each image has its own field. Leave it empty to keep the image, or describe the new image you want, for example *same dress in red, studio lighting*.
3. Click **Apply Feedback**. The app regenerates only the images you described and edits the text. The revised article is saved as a new file ending in `_revised.html`.

Edit an image with object detection

Each image also has an **Edit Image** button. It opens an editor that finds objects in the image (for example a person or a handbag) with the YOLO object detector and changes only that object.

1. Click **Edit Image** below the image you want to change.
2. Click **Run Detection**. After a few seconds, the detected objects are listed.
3. Click the object you want to change.
4. Choose an operation: **Gaussian Blur** softens the object (set the strength with the **Blur Radius** slider), **Inpaint** removes it and fills the area from the background, and **Make White** replaces it with white.
5. If you do not like the result, click **Undo Last Edit**.
6. Click **Use in Article** to put the edited image into the article, or **Discard** to close the editor without changes.

9 Stop the app and start it again later

Stop the app

Click into the terminal window that runs the app and press **Ctrl + C** (on macOS as well, not Cmd + C). The prompt comes back and the page in the browser stops working.

Start the app again

You do not need to repeat the installation. Open a new terminal and run these lines one at a time:

macOS

```
cd ~/w3-project-public/fashion_app
source venv/bin/activate
cd backend
python main.py
```

Windows

```
cd $HOME\w3-project-public\fashion_app
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass -Force
.\venv\Scripts\Activate.ps1
cd backend
python main.py
```

Then open **http://127.0.0.1:5001** in your browser and save your API key again, unless you created the `.env` file.

10 Troubleshooting

Find the message you see in the left column. If your problem is not listed, copy the last lines of the terminal and show them to a Teaching Assistant.

Problem or message	Solution
python3.13: command not found (macOS)	Python 3.13 is not installed, or the terminal was opened before the installation. Install it as in Step 2, close the terminal with Cmd + Q and open a new one.
'py' is not recognized, or No suitable Python runtime found (Windows)	Python 3.13 is not installed. Install it as in Step 2, then close PowerShell and open a new window.
running scripts is disabled on this system (Windows)	Run <code>Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass -Force</code> in the same window, then activate the environment again.
(venv) does not appear in the prompt	The environment is not active. Go to the <code>fashion_app</code> folder and run the activation command from Step 4.
No module named 'flask' (or another module)	The app was started without the active environment, or Step 5 did not finish. Activate the environment, run Step 5 again, then start the app.
No matching distribution found for torch==2.10.0	Wrong Python version or an Intel Mac. Inside the environment, <code>python --version</code> must show 3.13. Delete the <code>venv</code> folder and repeat Steps 4 and 5 with Python 3.13.

Problem or message	Solution
CERTIFICATE_VERIFY_FAILED or SSL error (macOS)	Open Finder, go to Applications, open the folder Python 3.13 and double-click Install Certificates.command. Then start the app again.
Errors mentioning a long file path, or Filename too long (Windows)	Run the Python 3.13 installer again, choose Modify, click Next, and on the last screen click Disable path length limit. Then repeat Steps 4 and 5. Keeping the project in your home folder also keeps paths short.
can't open file 'main.py' or No such file or directory	You are not in the backend folder. Run the <code>cd</code> command from Step 6 first.
Address already in use, or Port 5001 is in use	The app is already running in another terminal window. Use that window, or stop it there with Ctrl + C. If you cannot find it, restart your computer.
The browser shows This site can't be reached	The app is not running or still loading. Check the terminal and wait for Running on <code>http://127.0.0.1:5001</code> .
OpenRouter rejected this key	The key was mistyped or is no longer valid. Paste it again without spaces before or after it.
No OpenRouter API key set	Enter your key in the sidebar and click Save Key (Step 7). If you use a <code>.env</code> file, check that it is in fashion_app and not in backend .
Error code 402 or Insufficient credits	Your OpenRouter account has no credits left. Contact a Teaching Assistant.
Generation stops with a timeout or network error	Check your internet connection, wait a minute and click Generate again.
The first start seems to hang	On the first start the app downloads about 700 MB of models. Wait up to 5 minutes before you stop it.

Quick reference

Project folder: `w3-project-public/fashion_app` in your home folder.

App address: `http://127.0.0.1:5001`

Start: activate the environment, go to [backend](#), run `python main.py`.

Stop: Ctrl + C in the terminal that runs the app.

Your results: `fashion_app/output`